XO-1b

R-0054

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_XO-1_250602` · created 2026-07-10T04:39:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460828.7394 +/- 0.0067 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.053
Transit depth (Rp/Rs)^2	2.82 +/- 1.78 [%]
Orbital Inclination (inc)	84.59 +/- 1.81
Airmass coefficient 1 (a1)	1.32 +/- 0.76
Airmass coefficient 2 (a2)	-0.23 +/- 0.98
Scatter in the residuals of the lightcurve fit is	59.16 %
Best Comparison Star	None
Optimal Aperture	2.69
Optimal Annulus	8.81
Transit Duration (day)	0.05 +/- 0.047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.168 ± 0.053 vs 0.1326 (deviation 0.67σ)
Duration fit vs published	0.05 d vs 0.1238 d (deviation 1.57σ)
Mid-time O–C	2.2 min
Depth SNR	3.2
Residual scatter	59.16 %

Light curve

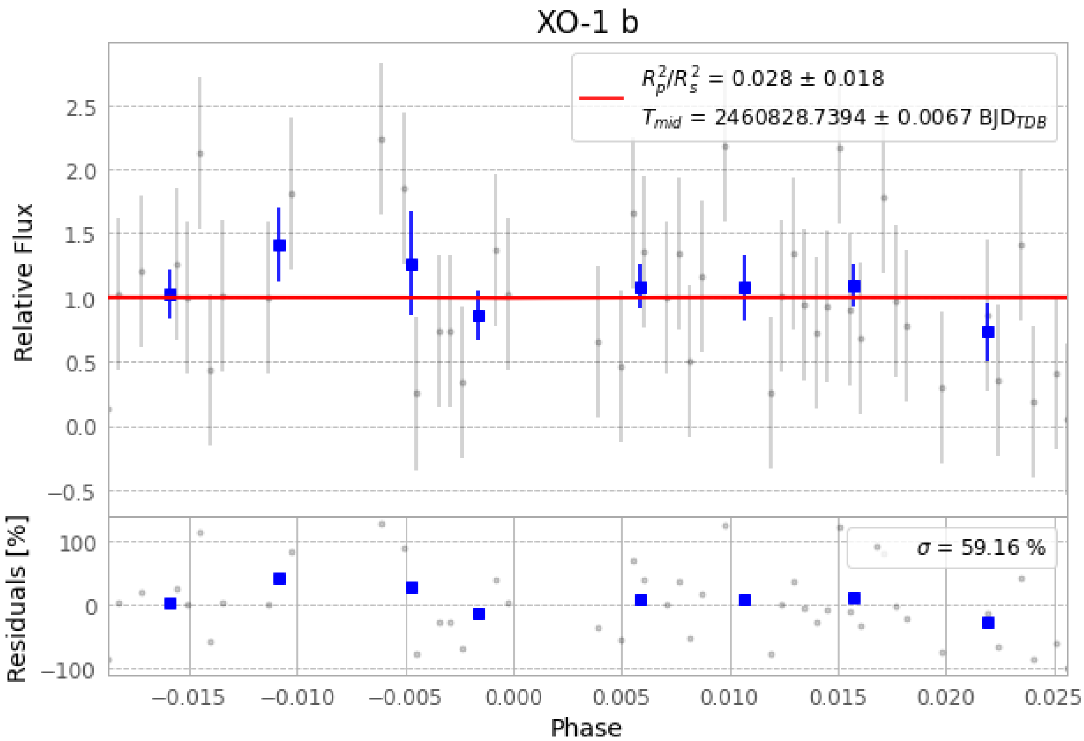

Figure 1 — FinalLightCurve_XO-1 b_2025-06-01.png (SHA-256 d48e04a712f7ac2b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 62], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 305fec9831579ec3…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

88 FITS frames (58 MB), `XO-1250602034215.FITS` → `XO-1250602080315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-250602.log (4e9d3bc5d6ae…), FinalLightCurve_XO-1 b_2025-06-01.png (d48e04a712f7…), FinalLightCurve_XO-1 b_2025-06-01.csv (027072afaf85…)

Compute resources

Wall time 139.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-15 16:40Z.